TI TIDA-060029 analog front end

Matched RG/RF: 100R / 5k / 100k

File: afe.kicad_sch

DDS excitation and reconstruction

100Hz-100kHz small-signal excitation

File: source.kicad_sch

Simultaneous voltage / current acquisition

Two channels / one conversion instant

File: adc.kicad_sch

USB, RP2040 and boot / debug

USB control, firmware and host calibration

File: digital.kicad_sch

Power: 5V input, forced PWM, quiet analog rails

External 5V → quiet +/-12V

File: power.kicad_sch

Engineering prototype. One PCB. Isolated, discharged DUT only.

Temper

Sheet: /
File: temper-lcr.kicad_sch

Title: Temper LCR — TIDA-060029 based

Size: A3	Date: 2026-10-20	**Rev: D**
KiCad E.D.A. 10.0.4		Id: 6/6

U6A source buffer; U6B HI sense; U7A transimpedance; U7B LO sense.
Fit ONE MATCHED pair: JP1+JP4=100R; JP2+JP5=5k; JP3+JP6=100k.
Power OFF to change ranges. One local C104=100pF compensates all ranges.
No BUF634A: maximum source current approximately 18mA. Isolated, discharged DUT only.

TI source and range network

JP1
R40
RG 100R
JP2
100R 0.1%
R41
RG 5k
JP3
5k 0.1%
R42
RG 100k
100k 0.1%
U6A
EXCITE
OUT
OPA2810IDGKR
SOURCE_BUF

TI transimpedance topology; Rev C compensation
C104=100pF directly across U7.1 and U7.2 in all ranges

R43
JP4
100R 0.1%
R44
RF 100R
JP5
5k 0.1%
R45
RF 5k
JP6
100k 0.1%
RF 100k
C104
100pF C0G 5%
U7A
GND
OUT
OPA2810IDGKR
TIA_OUT

Rev D: R94 22R 0.1% between J3 LO and the TIA summing node.
LOW follower senses J3 LO; host uses Zf = R94 + RF||C104.

J3
DUT_HI
DUT_LO
DUT HI / LO
R94
22R 0.1%
TIA_SUM

J3: connect external, isolated DUT between HI and LO.
No DUT component is fitted on this PCB.

U6B
HIGH_BUF
OPA2810IDGKR

U7B
LOW_BUF
OPA2810IDGKR

Independent HI / LO voltage followers

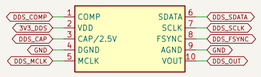
U6C
C42
100nF 50V
C44
1uF 50V
C50
10nF 50V
C43
100nF 50V
C45
1uF 50V
C51
10nF 50V
OPA2810IDGKR

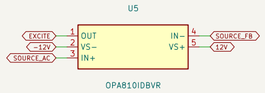
U7C
C46
100nF 50V
C48
1uF 50V
C52
10nF 50V
C47
100nF 50V
C49
1uF 50V
C53
10nF 50V
OPA2810IDGKR

Retain TI 100nF + 1uF bypass per rail; added 10nF at each supply pin.
PCB plane windows and short feedback routes are specified in reference-conformance.md.

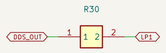
Temper	
Sheet: /TI TIDA-060029 analog front end/	
File: afe.kicad_sch	
Title: TI TIDA-060029 analog front end	
Size: A2	Date: 2026-10-20
KiCad E.D.A. 10.0.4	Rev: D
	Id: 2/6

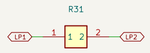
24MHz DDS clock from RP2040 120MHz / 5: sine output only.
Two 330R/1nF RC stages, AC coupling, OPA810 noninverting gain 6.
This block is added to the TI reference. Measure actual amplitude and distortion.

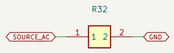


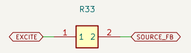


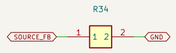














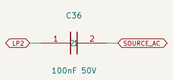




















Global net names connect functional sheets. See TI conformance matrix.

Temper

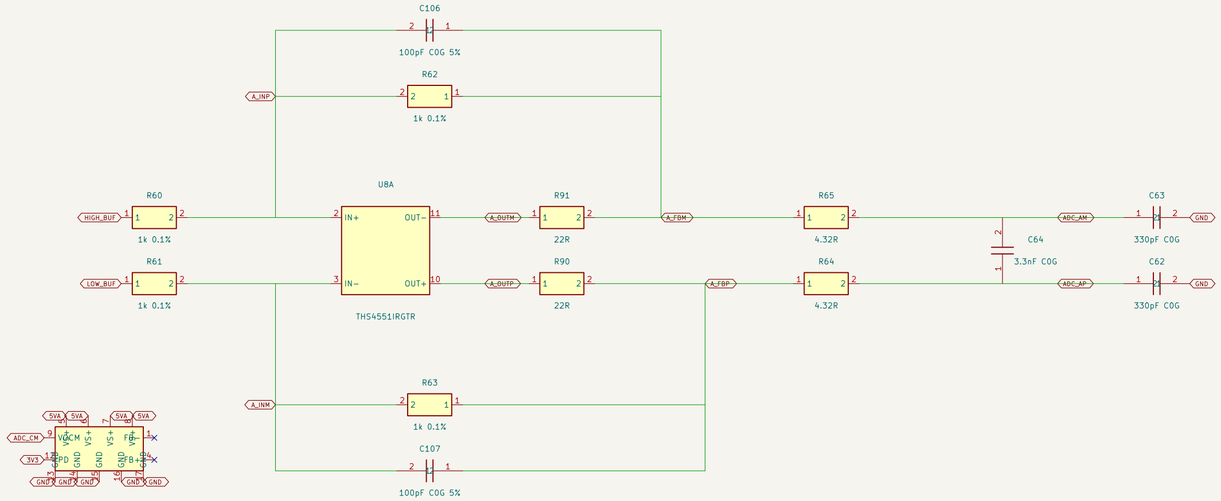
Sheet: /DDS excitation and reconstruction/
File: source.kicad_sch

Title: DDS excitation and reconstruction

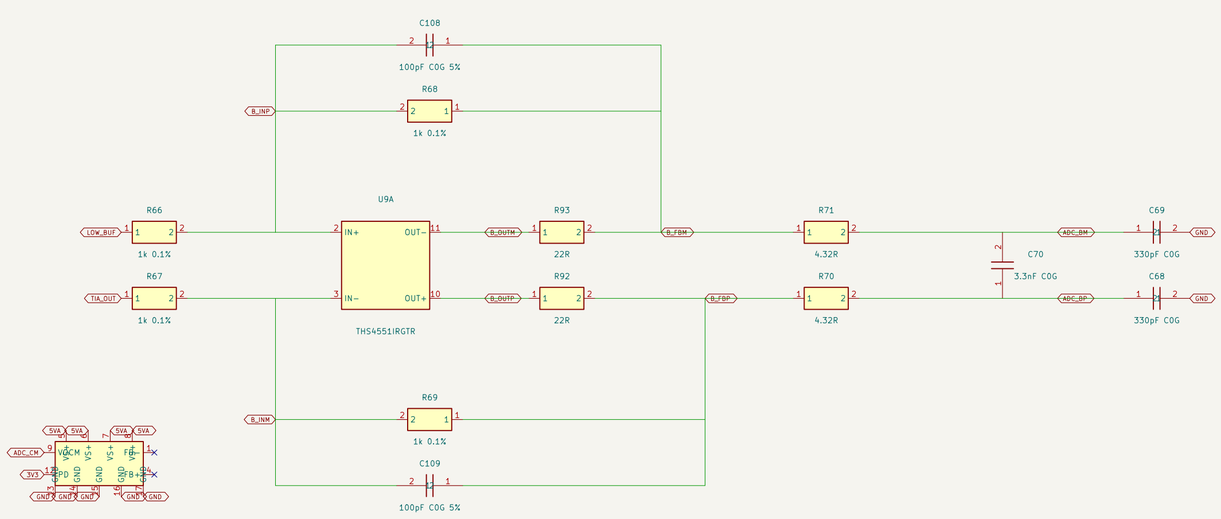
Size: A2	Date: 2026-10-20	**Rev: D**
KiCad E.D.A. 10.0.4		Id: 3/6

Channel A = HIGH_BUF – LOW_BUF; channel B = LOW_BUF – TIA_OUT.
Z = [RF / (1+j*2*pi*f*RF*CF)] * A/B. CF=100pF on every range; high-range pole=15.9kHz.
ADS9224REVM p26 topology: 22R inside feedback, 100pF || 1k, then 4.32R to ADC.
REFby2 drives VOCM. REFOUT is decoupled only. Each REFP uses 22uF.

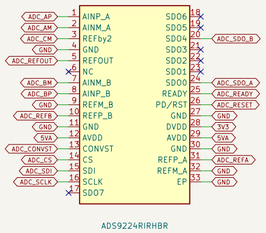
U8: feedback senses the nodes AFTER 22R, BEFORE 4.32R.



U9: feedback senses the nodes AFTER 22R, BEFORE 4.32R.

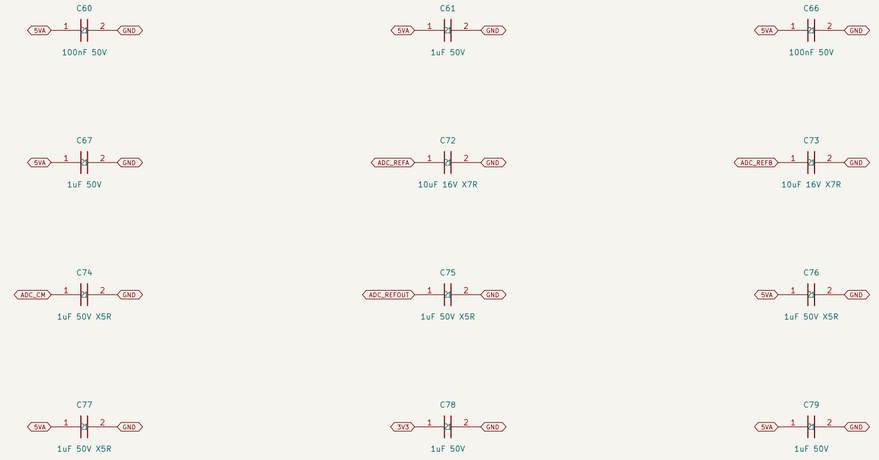


U10



ADS9224RIRHBR

Reference and supply bypass. REFby2 directly drives both VOCM pins;
C74=1uF. EVM per-driver 100R / 100nF VOCM filters are omitted.
RGT internal FB pads are NC; external feedback uses main OUT pads.



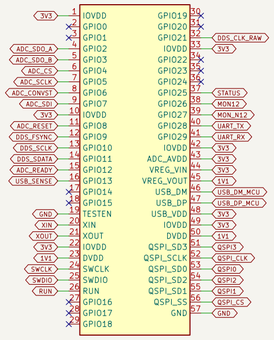
Temper
Sheet: /Simultaneous voltage / current acquisition/
File: adc.kicad_sch

Title: Simultaneous voltage / current acquisition

Size: A2	Date: 2026-10-20	**Rev: D**
KiCad E.D.A. 10.0.4		Id: 4/6

External 5V input powers the instrument. USB VBUS is sensed only; no backfeed.
Native USB CDC; 120MHz MCU; dual-lane ADC acquisition at up to 1MSPS.
Hold BOOTSEL while resetting for ROM USB firmware loading. SWD and UART provided.

U1



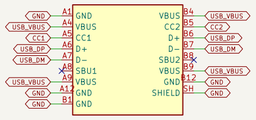
RP2040

U2



W25Q128JVSIQ

J1



USB-C

D1



USBLC6-2SC6

Y1

12MHz CL10pF

SW1

BOOTSEL

SW2

RESET

D4

RED

J4

SWD CLK / DATA

J5

SWD GND / VTREF

J6

UART TX / RX

R1

5.1k

R2

5.1k

R3

27R

R4

27R

R5



100k

R6

100k

R7

10k

R8

1k

R9

10k

R10

1k

R11

33R

R12

1k

R13

33R

C1

100nF 50V

C2

100nF 50V

C3

100nF 50V

C4

100nF 50V

C5

1uF 50V X5R

C6

100nF 50V

C7

100nF 50V

C8

1uF 50V

C9

100nF 50V

C10

1uF 50V

C11

100nF 50V

C12

15pF C0G

C13

15pF C0G

C14

100nF 50V

C15

100nF 50V

R98

10k

R99

10k

C110

100nF 50V

C111

100nF 50V

Global net names connect functional sheets. See TI conformance matrix.

Temper

Sheet: /USB, RP2040 and boot / debug/
File: digital.kicad_sch

Title: USB, RP2040 and boot / debug

Size: A2	Date: 2026-10-20	**Rev: D**
KiCad E.D.A. 10.0.4		Id: 5/6

Input: REGULATED 5V +/−5%, 1A, center positive. USB does not supply this board.
TPS65131 PSP/PSN grounded: forced PWM 1.25−1.5MHz. RAW_P ~14.31V: RAW_N ~−14.39V.
TPS7A49 ~+11.97V: TPS7A30 ~−12.00V. Fixed output values are nominal.
Do not substitute capacitor voltage/package without checking DC−bias derating.

U3
3.3V AP2112K

U11
TPS7A4901DGNR

U12
TPS7A3001DGNR

U13
TPS65131RGER

Q1
AO3401A

D2
SS14

D3
SS14

L1
4.7uH 3.8A shielded

L2
4.7uH 3.8A shielded

FB1
100R@100MHz 800mA

J2
5V 1A CENTER+

R95
1R

R96
1R

R80
100R

R81
1.62M

R82
150k

R83
1.74M

R84
150k

R85
91k 0.1%

R86
10k 0.1%

R87
91k 0.1%

R88
1k 0.1%

R89
10k 0.1%

C16
1uF 50V

C17
1uF 50V

C18
10uF 25V X5R

C80
10uF 50V X7R

C81
100nF 50V

C82
10uF 50V X7R

C83
100nF 50V

C84
220nF 25V

C85
4.7nF 50V

C86
10nF 50V

C87
4.3pF C0G

C88
4.3pF C0G

C89
10uF 50V X7R

C90
10uF 50V X7R

C91
10uF 50V X7R

C92
10uF 50V X7R

C93
10nF 50V

C94
10uF 50V X7R

C95
10uF 50V X7R

C96
10nF 50V

C97
100nF 50V

C98
10nF 50V

C99
10uF 50V X7R

C100
10uF 50V X7R

C101
10nF 50V

C102
100nF 50V

C103
22uF 25V X5R

R97
100k

R100
150k

PWR_FLAG
PWR_FLAG
PWR_FLAG
PWR_FLAG
PWR_FLAG
PWR_FLAG
PWR_FLAG
PWR_FLAG
PWR_FLAG
PWR_FLAG

Global net names connect functional sheets. See TI conformance matrix.

Temper	
Sheet: /Power: 5V input, forced PWM, quiet analog rails/	
File: power.kicad_sch	
Title: Power: 5V input, forced PWM, quiet analog rails	
Size: A2	Date: 2026-10-20
KiCad E.D.A. 10.0.4	**Rev: D**
	Id: 6/6